

Yaroslav Prytula

Computer Vision Research Engineer

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Profile

Computer Vision Research Engineer with 5 years of experience in applied computer vision research. My experience covers end-to-end research and development, from data collection and experimental design to model development and evaluation, with first-author work accepted at CVPR 2025 and BMVC 2026. I am also a PhD student in Computer Science at the University of Tartu and Ukrainian Catholic University, focusing on machine learning and computer vision. At the Biomedical Computer Vision Group, I research transformer-based segmentation, detection, representation learning, and foundation models for medical imaging.

Skills

- Python, C++, PyTorch, PyTorch Lightning, Detectron2, MMDetection, NumPy, SciPy, pandas, scikit-learn, OpenCV
- Machine Learning, Object Detection, Instance Segmentation, Vision Transformers, Representation Learning, Self-Supervised Learning, Contrastive Learning, Experimental Design, Model Evaluation
- FastAPI, REST API, Flask, PostgreSQL, Docker, Terraform, AWS, Kubernetes, Linux, Bash, Git, CMake, gRPC, SLURM, Weights & Biases, Codex, Claude Code

Work Experience

Computer Vision Research Engineer, BCVL Research Group, Tartu Mar 2022 – present

Develop foundation models for medical imaging, focusing on representation learning and feature extraction for dense prediction tasks using transformer models and self-supervised post-training.

- Developed QCell and IAUNet, transformer-based biomedical instance-segmentation models, labeled and open-sourced instance-segmentation datasets Revvity-25 and Organoids, resulting papers accepted at BMVC 2026 and CVPR 2025.
- Researching self-supervised post-training of vision foundation models for microscopy representation learning and dense prediction.

Data Scientist, Ukrainian Catholic University ML Lab – SoftServe

Sept 2021 – May 2024

In collaboration with SoftServe's R&D department, I developed a video-based system for real-time vital-sign monitoring. The system analyzes the frequency domain of signals extracted from facial regions of interest detected and tracked using MediaPipe. I was responsible for developing a robust algorithm for detecting and tracking heart rate in real-time video. Specifically, designed and implemented a PID-inspired, stateful coarse-to-fine PSD peak-tracking algorithm for robust heart-rate estimation from rPPG signals. I was also responsible for designing, testing, and conducting data collection under varying illumination conditions. I also explored unsupervised computer vision and signal-processing approaches to improve robustness under changing illumination and other environmental conditions.

Publications

IAUNet: Instance-Aware U-Net

Yaroslav Prytula, Illia Tsioporenko, Ali Zeynali, Dmytro Fishman

Accepted for oral presentation at the **CVMI Workshop at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025)**.

QCell: Recombining and Aligning Cell Queries for Overlapping Instance Segmentation

Yaroslav Prytula, Anton Popov, Dmytro Fishman

Accepted at the **British Machine Vision Conference (BMVC 2026)**.

Education

PhD Researcher, Mathematics and Computer Science

Sept 2025 – present

Ukrainian Catholic University and University of Tartu

MSc, Computer Science, University of Tartu

Sept 2023 – June 2025

- GPA: 4.7/5.0

BSc, Computer Science, Ukrainian Catholic University

Aug 2019 – June 2023

Physics and Mathematics, Lviv Physics and Mathematics Lyceum

Aug 2015 – May 2019

Teaching Experience

ML Lecturer, Ukrainian Catholic University

Sept 2024 – present

I teach the Machine Learning, Deep Learning, and Computer Vision modules of the Artificial Intelligence course and supervise student research projects in Applied AI, Machine Learning, and Computer Vision.

Co-instructing the Master's course Deep Learning for Computer Vision at the University of Tartu. Delivered lectures on Vision Transformers for classification, segmentation, and object detection.

Scholarships / Awards

- Best MSc Thesis Award, University of Tartu (2025)
- Research Stipend (2025)
- Tuition Waiver (2025)
- Achievement Stipend (2024)

Selected Projects & Competitions

Handwritten to Data – Ukrainian Handwritten Document Recognition	2026
National AI challenge focused on converting Ukrainian handwritten documents into structured digital data. Developed a three-stage pipeline combining PP-DocLayoutV3 for layout detection, type-aware transcription with fine-tuned TrOCR and Qwen3-VL, and rule- and VLM-based post-processing.	
- 4th place overall	
HuBMAP + HPA – Hacking the Human Body	2022
Kaggle competition on identifying and segmenting functional tissue units (FTUs) across five human organs.	
- (26th place) (top 3%)	
Sartorius – Cell Instance Segmentation	2021
Kaggle competition on detection and segmentation of neuronal cell instances in microscopy images.	
- (143rd place) (top 10%)	
TensorFlow – Help Protect the Great Barrier Reef	2022
Kaggle competition on crown-of-thorns starfish detection and tracking underwater.	
- (26th place) (top 1%)	
RSNA-MICCAI Brain Tumor Radiogenomic Classification	2021
Kaggle competition on predicting a probability for the target of a malignant tumor in the brain (glioblastoma).	

Talks & Lectures

Computer Vision in Agriculture

Data Science Summer School, Ukrainian Catholic University, summer 2025/26.

Multimodal Generative Models in CV – GenAI School, Ukrainian Catholic University, 2025

Covered diffusion-based text-to-image generation models including DALL-E, Stable Diffusion, and FLUX. Focused on architecture and core concepts.

Zero-Shot Models in Computer Vision – Ukrainian Catholic University, 2025

Lecture on zero-shot models covering CLIP, SAM, FastSAM, and SAM2, Grounding DINO, and Grounded SAM for tracking.

Certificates

Oxford Machine Learning School	May 2025 – Aug 2025
OxML@2025. Oxford Machine Learning School. Oxford's Deep Medicine & CIFAR. MLx Generative AI. Representation Learning & Gen.AI	
Lviv Data Science Summer School – Ukrainian Catholic University	Aug 2024
Lviv Data Science Summer School (LDSSS@2024)	
Improving Deep Neural Networks	July 2021 – Aug 2021
Coursera course on Deep Learning	
IELTS	Mar 2023
7.5 Overall Band Score (C1 CEFR Level)	

Languages

English: Full Professional Proficiency (C1)

Ukrainian: Native or Bilingual Proficiency